

NMAM INSTITUTE OF TECHNOLOGY, NITTE*Off-Campus Centre of Nitte (Deemed to be University)***Third Semester B. Tech (CBCS) Degree Examinations**

Summer Semester July 2024

EE2104-1 – GENERATION TRANSMISSION AND DISTRIBUTION

(EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- The commercial sources of energy are
 - Fossil fuels, water and radioactive substances
 - Solar, wind, biomass
 - Fuel cells
 - All of these
- A hydroelectric power station is commonly found in
 - desert areas
 - hilly areas
 - swamps
 - Grasslands
- A penstock is used as a conduit between
 - The steam chest and the turbine in a thermal station
 - The dam and the turbine in a hydro station
 - The turbine and the discharge drain
 - The turbine and the tail race
- Running cost of power plant is based on the cost of
 - Energy or fuel
 - Consumable items
 - Maintenance and operation
 - All of these
- All types of tariffs must recover the cost of
 - Capital investment in generating equipment
 - Operation, supplies, and maintenance of equipment
 - Metering equipment, billing, collection costs, profit and wages
 - All of these
- Factors affecting CORONA loss are _____
 - Frequency
 - Load current
 - System voltage
 - All of these
- Transmission and distribution of electric power by underground system is superior to overhead system in respect of-
 - Appearance and public safety
 - Maintenance cost
 - Frequency of faults, power failure and accidents
 - All of the above
- In a transmission system the feeder supplies power to-
 - Transformer substations
 - Service mains
 - Distributors
 - All
- The sag of a transmission line depends upon
 - Weight of conductor and directly proportional to span length
 - Weight of conductor and inversely proportional to span length
 - Weight of conductor and directly proportional to square of span length
 - Weight of conductor and inversely proportional to square of span length
- Effect of temperature rise in overhead lines is to
 - increase the sag and decrease the tension
 - decrease the sag and increase the tension
 - increase both the sag and tension
 - decrease both the sag and tension

11. 8. Which of the following material is not used for overhead line insulators?
 A) Porcelain B) Glass
 C) PVC D) Steatite
12. Why are arcing horns used ?
 A) to protect the towers B) to protect the insulators in HV transmission lines
 C) to protect the cables D) to protect the loads
13. The inductances of the transmission line do not contain effect of earth electric field geometry while the capacitances contain.
 A) True B) False
14. In the nominal pi method, which among these are divided into two halves?
 A) Series impedance B) Shunt capacitance
 C) Both (A) and (B) D) Neither A) nor B)
15. What happens in a long transmission line under no load?
 A) The receiving end voltage is less than the sending end voltage. B) The sending end voltage is less than receiving end voltage
 C) The sending end voltage is equal to receiving end voltage. D) None of these
16. When does the Ferranti effect happen on the transmission line?
 A) When the line is short and loaded. B) When the line is long and loaded.
 C) When the line is long and unloaded. D) None of these
17. What does capacitance grading of cables mean?
 A) Use of dielectrics in different concentrations. B) Introduction of capacitance at various lengths of cable to counter the effect of inductance.
 C) Use of dielectrics of different permittivity. D) Grading according to capacitance per km length of the cable.
18. What is the main type of distribution system in India?
 A) Radial B) Parallel
 C) Network D) Both (b) and (c)
19. For purely domestic loads which type of distribution is employed?
 A) Single phase two wire. B) 3 phase 3 wire
 C) 3 phase 4 wire D) None of these
20. FACTS devices are generally used to compensate -----of the transmission line
 A) Reactance B) Resistance
 C) Conductance D) Admittance

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the working of diesel power plant with details of all the components, & a neat sketch.	4	L*2	1	1,2
	b) Define the following: - i) Load Factor ii) Plant Capacity Factor iii) Diversity Factor iv) Demand Factor v) Utilization Factor vi) Maximum Demand	6	L1	2	1,2
	c) With neat schematic explain the working of thermal power plant with brief description of every main part.	6	L2	1	1,2
2.	a) Explain the following: - i) Penstock ii) Dam iii) Surge tank	4	L1	1	1,2

- b) Maximum demand of a generating station is 100MW with a load factor of 65%. The plant capacity factor & plant use factors are 50% & 80% respectively. Determine: -
- Daily Energy produced.
 - Installed plant capacity.
 - Reserve capacity of plant
 - Maximum energy that could be produced daily if the plant is running all the time.
 - Maximum energy that could be produced daily if the plant runs at full load.
 - Utilization Factor
- c) With a neat diagram illustrate the essential components of nuclear reactor.
3. a) Define tariff. Explain different types of tariffs.
- b) A station supplies the following loads: - 15,000kW, 12,000kW, 8,500kW, 6000kW & 450kW. The station has a maximum demand of 22000kW. The annual load factor of the station is 48%. Calculate
- Number of units supplied annually.
 - Diversity factor
 - Demand factor
- c) Draw the schematic diagram of Gas turbine power plant & explain its working.
- Unit – II**
4. a) Derive an expression for sag of a transmission line when suspended on supports at equal level.
- b) Write a note on Corona in transmission lines. Also explain the methods to reduce corona.
- c) Explain Ferranti effects in transmission lines.
5. a) A transmission line has a span of 200m between level supports. The conductor has a cross sectional area of 1.29 sq.cm, weighs 1170kg/km and has breaking stress of 4218 kg/sq.cm. Calculate the sag. Allow wind pressure of 122 kg/sq.m of projected area. What is the vertical sag? Take factor of safety as 5.
- b) Derive the expression for sending end voltage and current of a medium transmission line. Use nominal-T method.
- c) Discuss the concept of self GMD and mutual GMD.
6. a) Derive the expression for inductance of a three-phase overhead line if conductors are symmetrically spaced.
- b) Calculate the sending end voltage and current of a medium transmission line Using nominal-pi method.
- c) Briefly explain the potential distribution over a string of suspension insulators.
- Unit – III**
7. a) Briefly describe the different types of cables used in underground systems
- b) A single core cable has a conductor with a diameter of 1cm and insulation thickness of 0.4cm. If the specific resistance of the insulation is 5×10^{14} ohm-cm, then find the insulation resistance for a 2km length of the cable.
- c) Derive an expression for the capacitance of a single core underground cable.
8. a) List the advantages and disadvantages of underground cables.
- b) Briefly explain the concept of radial distribution system with a neat sketch.
- c) A two-wire distribution system is 3km long and it supplies loads of 200A, 100A, 75A and 50A at 800m, 1200m, 2000m & 3000m respectively from feeding point A. Each conductor has go & return resistance of 0.004 ohm/100m. Calculate the voltages at each load point if the voltage at feeding point is 250V.

6	L3	2	1,2
6	L2	1	1,2
6	L1	2	1,2
4	L3	2	1,2
6	L1	1	1,2
6	L3	3	1,2
5	L2	4	1,2
5	L1	4	1,2
6	L3	4	1,2
4	L3	4	1,2
6	L2	3	1,2
6	L3	3	1,2
6	L3	4	1,2
4	L2	3	1,2
6	L2	5	1,2
4	L3	5	1,2
6	L2	5	1,2
5	L1	5	1,2
5	L2	5	1,2
6	L3	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
